



LLMs & VLMs course

Introduction to Large Language Models

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The A.I. revolution

- ChatGPT became publicly available in November 30 2022^a
- It reached 100 million users in just two months.
- 550 M in 2025.

^aGemini, Consulted May 26.

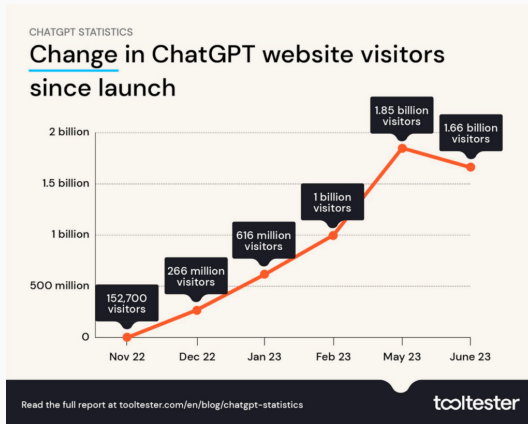


Figure 1:

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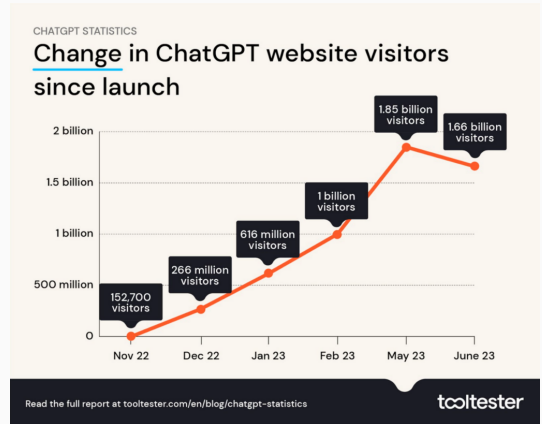


Figure 2:

- **Artificial Intelligence:** Systems capable of human-like intelligence tasks.
- **Machine Learning:** Algorithms that learn rules automatically from data (no explicit programming).
- **Deep Learning:** A subset of ML utilizing multi-layer neural networks.
- **Generative AI (GenAI):** Models designed to create new content (text, images, media).
- **LLMs:** Specific application of deep learning for parsing and generating human-like text.

The New Era of Natural Language Processing

- **Traditional NLP:** Excelled at categorization (spam detection) and pattern recognition via handcrafted rules or shallow models.
- **The LLM Shift:** Modern models (e.g., ChatGPT) leverage deep neural networks to manage complex linguistic tasks.
- **Capabilities:** Understanding instructions, contextual analysis, and coherent text generation.
- **Key Insight:** "Understanding" in LLMs refers to statistical coherence and contextual relevance, not human-like consciousness.

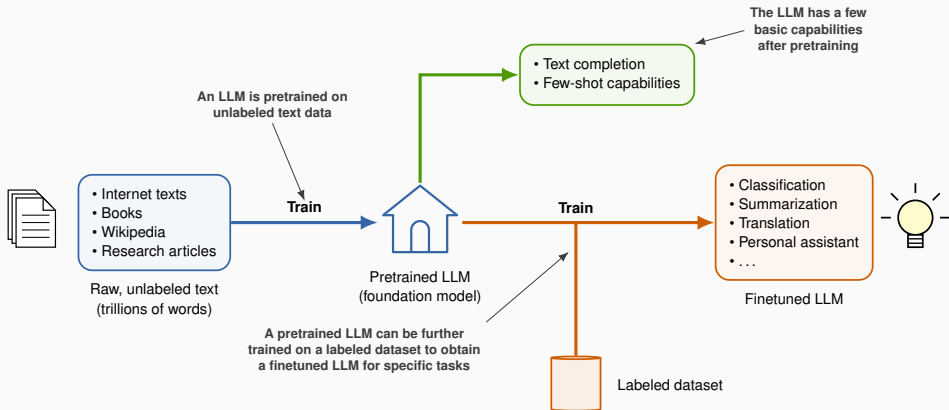
What is a Large Language Model (LLM)?

Definition

An LLM is a deep neural network designed to "understand", generate, and respond to human-like text by being trained on massive datasets.

- **"Large" dimensions:**
 - **Parameter Size:** Often tens or hundreds of billions of weights optimized during training.
 - **Dataset Scale:** Sometimes encompassing significant portions of the publicly available internet.
- **Core Objective:** Next-word prediction (leveraging the sequential nature of language).

Development



The Two-Stage Training Paradigm

LLM development typically follows a bifurcated process:

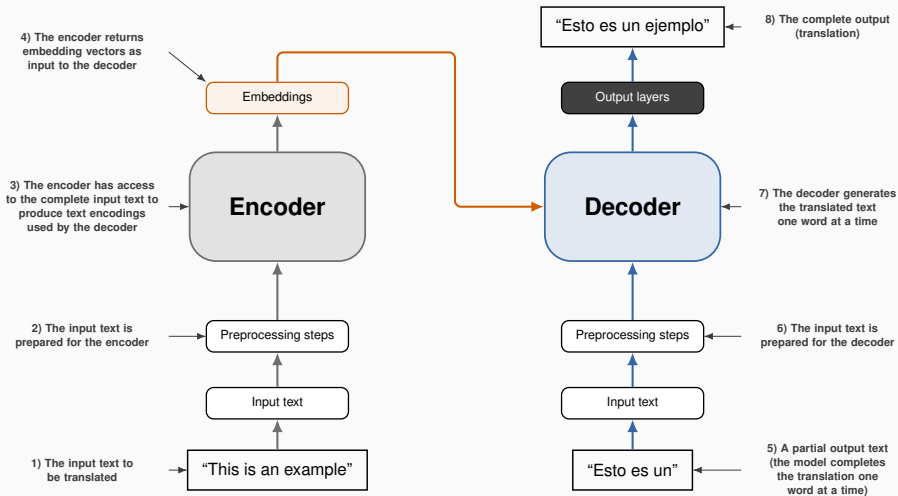
1. Pretraining (Self-Supervised):

- Trained on large, unlabeled corpora (trillions of words).
- Result: **Foundation/Base Model** (e.g., GPT-3).
- Capability: Text completion and few-shot learning.

2. Finetuning (Supervised):

- Narrow dataset specific to tasks or domains.
- **Instruction-finetuning:** Adapting for chat/assistants.
- **Classification-finetuning:** Specializing for sentiment or toxicity detection.

Transformer insight



The Transformer Foundation

Most modern LLMs rely on the architecture introduced in "*Attention Is All You Need*" (2017).

- **Encoder-Decoder:** Original design for machine translation (English → German).
- **Self-Attention Mechanism:** Allows the model to weigh the importance of different tokens in a sequence relative to each other, capturing long-range dependencies.
- **Architectural Variants:**
 - **BERT:** Encoder-only; specialized in masked word prediction and classification.
 - **GPT:** Decoder-only; specialized in generative tasks and autoregressive word prediction.

Scaling Laws: Data for LLMs

LLMs require diverse and comprehensive text corpora encompassing billions of words.

Table 1: GPT-3 Pretraining Dataset (Sampled 300B Tokens)

Dataset	Description	Proportion
Common Crawl	Filtered web crawl (410B tokens)	60%
WebText2	Outbound Reddit links (3+ upvotes)	22%
Books1/2	Project Gutenberg / Libgen	16%
Wikipedia	High-quality text (English)	3%

- **Cost:** GPT-3 pretraining estimated at \$4.6M in cloud credits.

Emergent Behaviors and Multitasking

- **Definition:** Capabilities not explicitly taught during training but arising as a natural consequence of large-scale data exposure.
- **Example:** GPT models can perform translation despite being trained primarily on next-word prediction.
- **Foundation Models:** Their versatile nature allows them to be used as general-purpose tools for tasks not seen in training data.
- **Advantage:** Custom-built LLMs (e.g., BloombergGPT) can outperform general LLMs in specific domains like finance.

Building an LLM from Scratch

The roadmap for the implementation:

- **Stage 1: Data and Architecture:** Data preparation, tokenization (BPE), and coding the Attention mechanism.
- **Stage 2: Pretraining:** Implementing the training loop and evaluating performance (Educational scale).
- **Stage 3: Finetuning:** Taking a base model and adapting it for specific instruction-following or classification.

- Raschka, Sebastian. Build a large language model (from scratch). Simon and Schuster, 2024.
- Goodfellow, Ian, et al. Deep learning. Vol. 1. No. 2. Cambridge: MIT press, 2016.

Questions?